Name ____ Date ___________ Period _____

Parent Functions III: 1/x and Asymptotes

Algebra II Honors | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, describe its end behavior precisely, prove it is odd, and locate vertical asymptotes of related rules from the denominator.

Vocabulary

Vertical asymptote — a vertical line the graph _________________, located where the function is _______________.

Horizontal asymptote — describes _________________. Unlike a vertical one, it ________ be crossed by other functions.

End behavior notation — "as x → ∞, f(x) → 0" is read: as x grows ________________, the output approaches 0.

Part 1 — Build the Table

x	-4	-2	-1	-1/2	0	1/2	1	2	4
f(x) = 1/x									

Why is x = 0 excluded, and what is the domain in set notation?

Division by zero is undefined, so ________________________

Part 2 — End Behavior, Precisely

Complete both patterns, then write the behavior in words.

x	1	0.1	0.01	0.001
1/x				

x	1	10	100	1000
1/x				

As x → 0 from the right, f(x) → ____________ As x → 0 from the left, f(x) → ____________

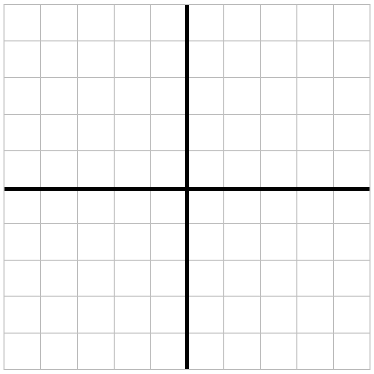
As x → ∞, f(x) → ____________ As x → -∞, f(x) → ____________

So the asymptotes are x = __________ and y = __________. Does the graph ever reach y = 0? __________

Part 3 — Graph and Prove

Each square is 1 unit. Two branches - do not connect them across the y-axis.

$f(x) = 1/x$, with both asymptotes dashed.



Prove f is odd.

$f(-x)$ = ____________________

so $f(-x)$ = ________________

Quadrants used: ________________

Intercepts: ________________

Max or min: ________________

Why does having no intercepts follow from the asymptotes?

____

The axes _________ the asymptotes, and a graph never reaches its asymptotes.

Part 4 — Asymptotes of Related Rules

Reason from the denominator. Where it is zero, there is a vertical asymptote.

Function	Denominator = 0 when	Vertical asymptote(s)	Domain
$1/(x - 3)$	_____________	_____________	____________
$1/(x^2 - 4)$	_____________	___________	____________
$1/(x^2 + 1)$	_____________	_____________	____________

The last row has no vertical asymptote at all. Explain why:

____

$x^2 + 1$ is __________________________, so it is never zero for a real x.

Part 5 — Going Further

1. True or false, with a reason: a graph may cross a horizontal asymptote. ________________
2. Does $1/x$ itself ever cross $y = 0$? ________________ Is that a contradiction with item 1?

____

3. Solve $1/x = x$. Then say what the solutions mean on the graph.

____

x = _____________________ — ______________________________________

Exit Ticket

Give the domain and both asymptotes of $f(x) = 1/(x + 5)$, and state its end behavior as $x \to \infty$.

Domain _____________________________ VA ________________ HA ________________

End behavior: as $x \to \infty$, $f(x) \to$ ________________